Customer Shopping Behavior Analysis

1. Project Overview

The project aims to analyze the shopping behavior dataset to identify sales performance, customer behavior, and analyze product performance, providing direction for business planning.

2. Dataset Summary

Rows: 9,900

Columns: 18

Key Features:

- Customer demographics (name, segment, country, city, state, postal code, region)
- Purchase details (category, sub-category)

Missing Data: 11 values in Postal Code column.

3. Exploratory Data Analysis (EDA) Using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` to check structure, `df.describe()` for summary statistics, and `df.isnull().sum` to check for missing values.

	row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment	country	city	state	postal_c
count	9800.000000	9800	9800	9800	9800	9800	9800	9800	9800	9800	9800	9800.000
unique	NaN	4922	1230	1326	4	793	793	3	1	529	49	I
top	NaN	CA-2018-100111	5/9/2017	26/09/2018	Standard Class	WB-21850	William Brown	Consumer	United States	New York City	California	I
freq	NaN	14	38	34	5859	35	35	5101	9800	891	1946	I
mean	4900.500000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	55241.83(
std	2829.160653	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	32037.012
min	1.000000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1040.000
25%	2450.750000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	23223.000
50%	4900.500000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	57551.000
75%	7350.250000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	90008.000
max	9800.000000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	99301.000

- **Missing Data Handling:** Checked for null values and imputed missing values in Postal Code column using the most frequent value (mode) within each city.
- **Data Cleaning:** Converted the date format in the DataFrame from DD/MM/YYYY to YYYY/MM/DD.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transaction)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by region –** Compared the total revenue generated by customers across four regions, with the West generating the highest revenue and the South generating the lowest.

	region text	total_revenue double precision
1	South	389151.4590000004
2	West	710219.6845000003
3	East	669518.7259999984
4	Central	492646.9132000005

2. **Revenue by City –** Compared total revenue generated by top 5 most selling city, with San Francisco generating the highest revenue.

	city text	total_revenue double precision
1	New York City	252462.547
2	Los Angeles	173420.181
3	Seattle	116106.322
4	San Francisco	109041.1199999999
5	Philadelphia	108841.7490000000

3. **Revenue by Category –** Compared total revenue generated by 3 categories, with Technology generating the highest revenue.

	category text	total_revenue double precision
1	Technology	827455.8729999964
2	Furniture	728658.5756999997
3	Office Supplies	705422.3340000035

4. **Shipping Type Comparison –** Compared average revenue between First Class, Second Class, and Standard Class shipping.

	ship_mode text	avg_revenue double precision
1	Standard Class	228.84985611878864
2	Second Class	236.54793869610933
3	First Class	230.22801952032

5. **Customer Segment Type Comparison –** Compared average revenue between Customer, Corporate, and Home Office

	segment text	avg_revenue double precision
1	Consumer	225.06577749460683
2	Corporate	233.1507195394515
3	Home Office	243.4033086483393

6. **Top 5 Products by Total Revenue –** Identified the 5 top-selling products within each category and subcategory.

	product_name text	category text	sub_category text	total_revenue double precision
1	Canon imageCLASS 2200 Advance...	Technology	Copiers	61599.824
2	Fellowes PB500 Electric Punch Plas...	Office Supplies	Binders	27453.384
3	Cisco TelePresence System EX90 V...	Technology	Machines	22638.48
4	HON 5400 Series Task Chairs for Bi...	Furniture	Chairs	21870.576
5	GBC DocuBind TL300 Electric Bindi...	Office Supplies	Binders	19823.479000000

7. **Total Revenue Each Year** – Identified the total revenue for each year of sales.

	year numeric	total_quantity bigint	total_revenue double precision
1	2015	1953	479856.208100001
2	2016	2055	459436.0054000001
3	2017	2534	600192.550000001
4	2018	3258	722052.0192000001

8. **Total Revenue Each Month in 2018 –** Identified 2018 as the year with the highest total revenue among the four years, and calculated total revenue by month for each year.

	month numeric	total_revenue double precision
1	11	117938.154999
2	9	86152.8879999
3	12	83030.3887999
4	10	77448.1312000
5	8	62837.8479999
6	3	58863.4127999
7	6	48190.7277
8	7	44825.1039999
9	5	43825.9821999
10	1	43476.4739999
11	4	35541.9100999
12	2	19920.9973999

9. **Top 5 Products by Revenue in November 2018** – Identified the 5 top-selling products within each category and subcategory in November 2018

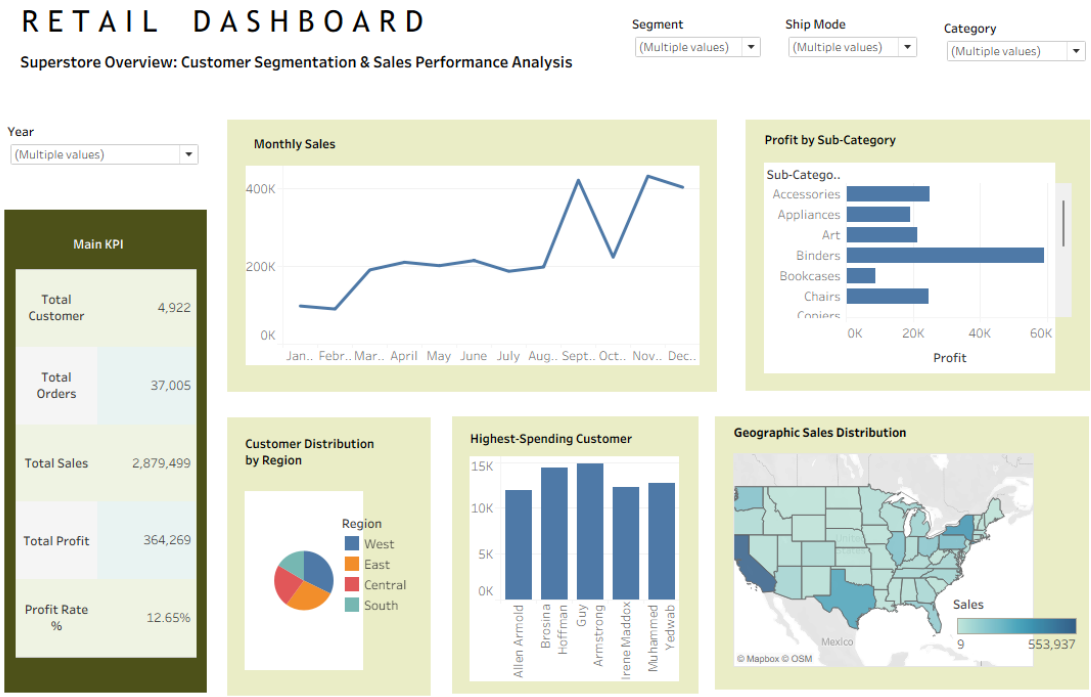
	product_name text	category text	sub_category text	total_revenue double precision
1	Canon imageCLASS 2200 Adva...	Technology	Copiers	10499.97
2	Cubify CubeX 3D Printer Triple ...	Technology	Machines	7999.98
3	Hon 94000 Series Round Tables	Furniture	Tables	2665.62
4	Hon 5100 Series Wood Tables	Furniture	Tables	2560.624
5	Hoover Shoulder Vac Commerci...	Office Supplies	Appliances	2504.74

10. **Lowest 5 Produts by Revenue in November 2018** – Identified the 5 lowest-selling products within each category and subcategory in 2018, November

	product_name text	category text	sub_category text	total_revenue double precision
1	Zipper Ring Binder Pockets	Office Supplies	Binders	1.248
2	Self-Adhesive Ring Binder Labels	Office Supplies	Binders	1.408
3	BIC Brite Liner Grip Highlighters	Office Supplies	Art	1.64
4	Avery Durable Binders	Office Supplies	Binders	2.88
5	Maxell 4.7GB DVD+R 5/Pack	Technology	Accessories	2.97

5. Dashboard in Tableau

Finally, an interactive dashboard was developed using Tableau to visualize the findings and provide a clearer overview of the retail business's performance.



6. Business Recommendation

- **Boost**

  Across the four-year period, sales peaked during the winter months, with the West region consistently recording the highest number of purchases for four consecutive years. In terms of product performance, the Furniture category generated the highest sales, particularly in the Chairs, Furnishings, and Bookcases sub-categories. Based on these findings, the business could consider introducing seasonal product collections or special-themed promotions to capitalize on the increased demand during the winter season.

- **Targeted Marketing**

  The Home Office segment generates the highest revenue, with Second Class being the most frequently used shipping method. Technology also stands out as the best-performing product category within this segment. Based on these findings, the business could focus on creating targeted marketing campaigns specifically for Home Office customers, highlighting popular Technology products through personalized promotions, bundle deals, and product recommendations. Additionally, offering shipping incentives

or discounts for Second Class deliveries could help encourage repeat purchases and strengthen customer loyalty.

- **Product Positioning**
  Considering the increase in sales during the winter months, the business could introduce seasonal product collections and limited-time promotions to capitalize on peak purchasing periods. Furniture and Technology products could be strategically marketed to align with seasonal customer needs. Furthermore, since the West region consistently generates the highest sales, the business should strengthen its market presence by prioritizing high-demand products in this area. Meanwhile, regions with lower sales performance, such as the South, could benefit from localized marketing campaigns and tailored product recommendations based on customer purchasing patterns to encourage engagement and improve overall sales performance.